Saturated fusion systems on groups of order 3^8

a running report

October 17, 2026

Abstract

We search for the saturated fusion systems $\mathcal{F}$ on the groups S of order 3^8 with $O_3(\mathcal{F}) = 1$ and $O^3(\mathcal{F}) = \mathcal{F}$, in the manner of Parker–Semeraro [1] but with the fast proto-essential search of Gautam [2] and the fusion-graph-free saturation test of the GAP package `FusionSystems` that accompanies this report. There are 1 396 077 groups of order 3^8 (Vaughan-Lee's database, available in GAP through the package `sglppow`). Section 3 reduces them to a short list by necessary conditions, Section 4 identifies the surviving groups, Section 5 contains the look-up tables of subgroups of $\mathrm{GL}_n(3)$ with a strongly 3-embedded subgroup, and Section 6 records the fusion systems found group by group. This document is updated as the computation proceeds; the current state is summarised in Section 1.

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1 Status

[updated as the work proceeds]

- Stage 1a (cheap reduction of the 1 396 077 groups): a first run started on 17 October 2026 was stopped after 17 600 groups when the 3-group test was found to use a wrong quantity (see Section 3); restarted the same day with the corrected pipeline on four cores, about 1.3 s per group, so roughly five days; about 2% of random groups survive it.
- Stage 1b (automorphism group tests on the survivors): not started.
- Look-up tables for $\mathrm{GL}_n(3)$: complete for $n \leq 4$, and for $n = 5$, $T \cong 3^2$ (Section 5).
- Stage 2 (fusion systems group by group): the results for 2221 (7 systems) and 2225 (1 system) are confirmed with the corrected pipeline; the other groups of Sections 6.3–6.4 will be recomputed as the restarted sweep reaches them.; the groups passing Stage 1b are treated in the order of the largest elementary abelian proto-essential subgroup.

2 Conventions and the strategy

Throughout $p = 3$, S is a group of order 3^8 and $\mathcal{F}$ is a saturated fusion system on S with $O_3(\mathcal{F}) = 1$ and $O^3(\mathcal{F}) = \mathcal{F}$. We write $\mathrm{Aut}_S(E) = N_S(E)/C_S(E)$ and $\mathrm{Out}_S(E) = \mathrm{Aut}_S(E)\mathrm{Inn}(E)/\mathrm{Inn}(E)$; for an S-centric subgroup E (that is $C_S(E) \leq E$) we have $\mathrm{Out}_S(E) \cong N_S(E)/E$.

By the Alperin–Goldschmidt fusion theorem $\mathcal{F}$ is generated by $\mathrm{Aut}_{\mathcal{F}}(S)$ and the groups $\mathrm{Aut}_{\mathcal{F}}(E)$ for the $\mathcal{F}$-essential subgroups E (the $\mathcal{F}$-centric, fully normalised subgroups with $\mathrm{Out}_{\mathcal{F}}(E)$ having a strongly 3-embedded subgroup). Following [1] we therefore proceed as follows.

1. Determine the subgroups $E \leq S$ which can be essential in some saturated fusion system on S (the *proto-essential* subgroups), up to $\mathrm{Aut}(S)$-conjugacy. Here we use Gautam's theorem (Theorem 2.1 below) instead of the subgroup lattice of S.
2. Determine the Borel subgroups, i.e. the $3'$-subgroups $K \leq \mathrm{Aut}(S)$ up to conjugacy, so that $\mathrm{Aut}_{\mathcal{F}}(S) = \mathrm{Inn}(S)K$ (recall that $\mathrm{Inn}(S)$ is a normal Sylow 3-subgroup of $\mathrm{Aut}_{\mathcal{F}}(S)$).
3. For each Borel subgroup and each proto-essential subgroup E determine the candidates for $\mathrm{Aut}_{\mathcal{F}}(E)$: the subgroups of $\mathrm{Aut}(E)$ containing $\mathrm{Inn}(E)$ with $\mathrm{Aut}_S(E)$ as a Sylow 3-subgroup, with Out having a strongly 3-embedded subgroup and compatible with the Borel subgroup.
4. For each choice of a set of essential subgroups (up to the action of $N_{\mathrm{Aut}(S)}(\mathrm{Aut}_{\mathcal{F}}(S))$) and automizers, build the fusion system, discard it unless $O^3(\mathcal{F}) = \mathcal{F}$, $O_3(\mathcal{F}) = 1$ and $\mathcal{F}$ is saturated, and finally remove isomorphic copies.

Steps 2–4 are only carried out for the groups surviving the reduction of Section 3. All computations are done with the GAP package `FusionSystems`; the scripts for this report are in the directory `order38/scripts` of the repository.

Theorem 2.1 (Gautam [2, Theorem 3.6])**.** *Let $S = L_1 > L_2 > \cdots > L_m = 1$ be a central series of S with all factors of order p which refines the lower central series. Every essential subgroup of a saturated fusion system on S is S-conjugate to a subgroup $C_S(xA/A) = \{s \in S : [s, x] \in A\}$ with $A = L_i \leq [S, S]$ and xA an element of order p of S/A.*

3 The reduction

3.1 Necessary conditions

We use the following necessary conditions on S. Conditions (R1)–(R3) involve only S, its candidate subgroups E and (for the few candidates passing the structural tests) $\mathrm{Aut}(E)$; they are applied to all 1 396 077 groups (Stage 1a). Conditions (R4)–(R6) need $\mathrm{Aut}(S)$ and are applied to the survivors (Stage 1b).

Lemma 3.1 ([1, Lemma 7.1])**.** (R1) *If $|S : \mathrm{Z}(S)| \le 9$ then S carries no saturated fusion system with $O_3(\mathcal{F}) = 1$.*

Lemma 3.2. (R2) *Let E be an essential subgroup of a saturated fusion system $\mathcal{F}$ on S and $T = \mathrm{Out}_S(E) \cong N_S(E)/E$. Then*

(a) E is S-centric and $E < N_S(E)$;

(b) T is a Sylow 3-subgroup of a group with a strongly 3-embedded subgroup, so T is cyclic, or elementary abelian of rank at least 2, or $T \cong 3^{1+2}_+$, or $T \cong \mathrm{Syl}_3(\mathrm{PSU}_3(9))$ of order 3^6;

(c) $|E/\Phi(E)| \ge |T|^2$;

(d) if $1 = D_0 < D_1 < \cdots < D_k = E$ is a chain of characteristic subgroups of E, then no element of $N_S(E) \setminus E$ centralises every factor D_i/D_{i-1};

(e) $\mathrm{Aut}(E)$ is not a 3-group, $\mathrm{Aut}_S(E) \cap O_3(\mathrm{Aut}(E)) = \mathrm{Inn}(E)$, and $\mathrm{Aut}(E)$ is insoluble unless T is cyclic;

(f) if T is not cyclic then $|\mathrm{Aut}(E)|$ is divisible by 5 for $T \cong 3^2$, by $7 \cdot 13$ for $T \cong 3^3$, by 7 for $T \cong 3^{1+2}_+$, by $5 \cdot 41$ for $T \cong 3^4$, by $11 \cdot 61$ for $T \cong 3^5$ and by $5 \cdot 73$ for $|T| = 3^6$.

Proof. (a) and (b) are part of the definition and the classification of the groups with a strongly p-embedded subgroup (for $p = 3$ and T non-cyclic the possibilities for $O^{3'}(\mathrm{Out}_{\mathcal{F}}(E)/O_{3'}(\mathrm{Out}_{\mathcal{F}}(E)))$ are $\mathrm{PSL}_2(3^n)$, $\mathrm{PSU}_3(3^n)$, ${}^2G_2(3^n)$, $\mathrm{PSL}_3(4)$ and M_{11}; the Ree groups have Sylow 3-subgroups of order at least 3^9). (c) is [1, Lemma 2.?]: $\mathrm{Out}_{\mathcal{F}}(E)$ acts faithfully on $E/\Phi(E)$ because the kernel of $\mathrm{Aut}(E) \to \mathrm{Aut}(E/\Phi(E))$ is a normal 3-subgroup and $O_3(\mathrm{Out}_{\mathcal{F}}(E)) = 1$ as E is $\mathcal{F}$-radical. (d) The automorphisms of E centralising all factors of a chain of characteristic subgroups form a normal subgroup K of $\mathrm{Aut}(E)$, and K is a 3-group (a $3'$-automorphism centralising all factors of a normal series of a 3-group is trivial). Hence $K\mathrm{Inn}(E)/\mathrm{Inn}(E) \le O_3(\mathrm{Out}(E))$ meets $\mathrm{Out}_{\mathcal{F}}(E)$ in a normal 3-subgroup, which is trivial, so $K \cap \mathrm{Aut}_S(E) \le \mathrm{Inn}(E)$, which is (d) as E is S-centric. (e) is the $\mathcal{F}$-radical condition $O_3(\mathrm{Out}_{\mathcal{F}}(E)) = 1$ together with the fact that $T \ne 1$ is not normal in $\mathrm{Out}_{\mathcal{F}}(E)$ (a group with a strongly 3-embedded subgroup has more than one Sylow 3-subgroup), so that $\mathrm{Out}(E)$ is not a 3-group; if T is not cyclic then $O^{3'}(\mathrm{Out}_{\mathcal{F}}(E)/O_{3'}(\mathrm{Out}_{\mathcal{F}}(E)))$ is a non-abelian simple group by the classification quoted in (b), so $\mathrm{Aut}(E)$ is insoluble. (f) That simple group is a section of $\mathrm{Out}(E)$ and is one of $\mathrm{PSL}_2(9)$, $\mathrm{PSL}_3(4)$, M_{11} for $T \cong 3^2$, $\mathrm{PSL}_2(27)$ for 3^3, $\mathrm{PSU}_3(3)$ for 3^{1+2}_+, $\mathrm{PSL}_2(81)$ for 3^4, $\mathrm{PSL}_2(243)$ for 3^5 and $\mathrm{PSL}_2(729)$ or $\mathrm{PSU}_3(9)$ for $|T| = 3^6$; the primes listed divide the orders of all groups in the respective list. □

In Stage 1a a subgroup E is called a *candidate* if it is S-conjugate to a subgroup $C_S(xA/A)$ of Theorem 2.1 and satisfies (a)–(f) of Lemma 3.2, where in (d) we use the chain $E > \Phi(E) > 1$ and then the covering pairs of the set of characteristic subgroups generated (by one round of intersections and products) by $\mathrm{Z}(E)$, $\mathrm{Z}_2(E)$, $\Phi(E)$, E', $\gamma_3(E)$, $\mho_1(E)$, $\Omega_1(\mathrm{Z}(E))$, $\mathrm{Z}(\Phi(E))$ and $C_E(E')$. (The

stability group of the set of covering pairs is contained in the stability group of any maximal chain in the set, so the argument of (d) applies.) The tests are applied in this order, so that $\mathrm{Aut}(E)$ (computed with the GAP package `autpgrp`) is only needed for the candidates surviving (a)–(d); the first part of (e) discards a good part of them, since many groups of order 3^7 have a 3-group of automorphisms. With `autpgrp` we computed the automorphism groups of all groups of order 3^k, $k \le 7$ (`data/aut3table.g`):

$\lvert E\rvert$	3^3	3^4	3^5	3^6	3^7
groups	5	15	67	504	9310
$\mathrm{Aut}(E)$ a 3-group	0	0	0	30	2119
$\mathrm{Aut}(E)$ insoluble	3	5	8	17	37

(A first version of this computation used `autpgrp`'s `glOrder`, which is 1 whenever the action of $\mathrm{Aut}(E)$ on $E/\Phi(E)$ is soluble, and so counted the groups with insoluble automorphism group; the sweep was restarted with the corrected criterion.) A candidate which is not elementary abelian must therefore be isomorphic to one of the groups with $|\mathrm{Aut}(E)|$ not a power of 3; this is tested first through a cheap isomorphism invariant (the orders of the centre, Frattini and derived subgroups, the exponent, the class, the abelian invariants, the upper and lower central series, $\mho_1$, $\Omega_1(\mathrm{Z})$ and the class sizes by element order; `data/fingerprints.g`), and $\mathrm{Aut}(E)$ is only computed for candidates with a matching fingerprint. Elementary abelian candidates $E \cong 3^n$ have $\mathrm{Aut}(E) = \mathrm{GL}_n(3)$, and for $n \le 4$ the class of $\mathrm{Aut}_S(E)$ in $\mathrm{GL}_n(3)$ must be one of the classes carrying a candidate automizer in the look-up tables of Section 5.

Lemma 3.3. (R3) *If $W \neq 1$ is a characteristic subgroup of S which is contained in and characteristic in every candidate, then $W \le O_3(\mathcal{F})$ for every saturated fusion system $\mathcal{F}$ on S.*

Proof. W is normalised by $\mathrm{Aut}_{\mathcal{F}}(S)$ and by $\mathrm{Aut}_{\mathcal{F}}(E)$ for every essential E (an essential subgroup is S-conjugate to a candidate E_0, and $W = W^s$ is characteristic in E_0^s). By Alperin–Goldschmidt W is normalised by every morphism of $\mathcal{F}$ defined on it, i.e. $W \trianglelefteq \mathcal{F}$. □

In Stage 1a "characteristic" is replaced by the sufficient condition of membership in the sets of visibly characteristic subgroups described above.

Lemma 3.4. *Let $\mathcal{E}$ be a set of representatives of the S-classes of the proto-essential subgroups of S: the candidates of Stage 1a, where a candidate whose $\mathrm{Aut}(S)$-orbit contains a subgroup which is not a candidate is discarded (all the conditions of Lemma 3.2 are $\mathrm{Aut}(S)$-invariant).*

(R4) If $\mathcal{E} = \emptyset$ then $O_3(\mathcal{F}) = S$ for every saturated $\mathcal{F}$ on S.

(R5) If a nontrivial characteristic subgroup of S is characteristic in every member of $\mathcal{E}$, then $O_3(\mathcal{F}) \neq 1$.

(R6) If $S \neq \langle [S, O^3(\mathrm{Aut}(S))],\ E : E \in \mathcal{E}\rangle$ then $O^3(\mathcal{F}) < \mathcal{F}$ for every saturated $\mathcal{F}$ on S.

Proof. (R4) and (R5) are as before. (R6): $O^3(\mathcal{F}) = \mathcal{F}$ if and only if the hyperfocal subgroup $\langle [P, O^3(\mathrm{Aut}_{\mathcal{F}}(P))] : P \le S\rangle$ is S, and by Alperin–Goldschmidt it is generated by $[S, O^3(\mathrm{Aut}_{\mathcal{F}}(S))]$ and the $[E, O^3(\mathrm{Aut}_{\mathcal{F}}(E))] \le E$ for E essential; finally $O^3(\mathrm{Aut}_{\mathcal{F}}(S)) \le O^3(\mathrm{Aut}(S))$. □

3.2 Statistics

[to be filled in when Stage 1a finishes]

4 The surviving groups

[to be filled in]

4.1 Sylow 3-subgroups of simple groups of order 3^8

The finite simple groups G with $|G|_3 = 3^8$ and non-abelian Sylow 3-subgroups are (up to the obvious isomorphisms):

- Alt(18), Alt(19), Alt(20) (and the symmetric groups), with $S \cong (C_3 \wr C_3) \times (C_3 \wr C_3)$;
- $\mathrm{PSp}_4(9)$, S a Sylow 3-subgroup of $\mathrm{Sp}_4(9)$;
- $\mathrm{PSL}_3(q)$ with $3^4 \parallel q-1$ (e.g. $q = 163$) and $\mathrm{PSU}_3(q)$ with $3^4 \parallel q+1$ (e.g. $q = 647$), with $S \cong \big(\{(x,y,z) \in C_{81}^3 : xyz = 1\} \rtimes C_3\big)/C_3$ of maximal class;
- ${}^3D_4(q)$ with $3^3 \parallel q^2-1$ (e.g. $q = 53$), where S is an extension of $C_{81} \times C_{27}$ by C_3 (not constructed here).

The groups $\mathrm{PSL}_2(3^8)$, $\mathrm{PSL}_2(q)$ with $3^8 \parallel q \pm 1$ and $\mathrm{PSp}_4(q)$ with $3^4 \parallel q^2-1$ have abelian Sylow 3-subgroups, which carry no fusion system with $O_3(\mathcal{F}) = 1$. No sporadic group has $|G|_3 = 3^8$. The first three groups are constructed in `scripts/known_groups.g`; their invariants are

S	$\lvert\mathrm{Z}(S)\rvert$	class	rank	exponent	$\lvert S'\rvert$	classes
$\mathrm{Syl}_3(\mathrm{Alt}(18))$	9	3	4	9	81	289
$\mathrm{Syl}_3(\mathrm{PSp}_4(9))$	9	3	4	9	81	161
$\mathrm{Syl}_3(\mathrm{PSL}_3(163))$	3	7	2	81	729	737

5 Look-up tables: subgroups of $\mathrm{GL}_n(3)$ with a strongly 3-embedded subgroup

If $E \cong 3^n$ is an elementary abelian essential subgroup then $\mathrm{Out}_{\mathcal{F}}(E) = \mathrm{Aut}_{\mathcal{F}}(E) \le \mathrm{GL}_n(3)$ has Sylow 3-subgroup $T = \mathrm{Aut}_S(E)$ of order at most 3^{8-n} (as E is centric and $N_S(E) > E$, in fact $n \le 7$) and a strongly 3-embedded subgroup. The following tables list, for each $\mathrm{GL}_n(3)$-class of admissible T (cyclic, elementary abelian or 3_+^{1+2}), the subgroups H with $T \in \mathrm{Syl}_3(H)$ and a strongly 3-embedded subgroup, up to conjugacy in $N_{\mathrm{GL}_n(3)}(T)$. For $n \le 4$ the tables are complete. They are produced by `scripts/gltables.g` and stored in machine-readable form in `data/gln_table.g`.

The classes of T are distinguished in the tables by $|N_{\mathrm{GL}_n(3)}(T)|$; when the same H appears several times for one T the copies are not conjugate under $N_{\mathrm{GL}_n(3)}(T)$ (they differ in the way H acts, e.g. in the determinant or in the composition factors). The column "factors" gives the dimensions of the composition factors of H on the natural module, so that e.g. $1+2$ for $\mathrm{SL}_2(3) \le \mathrm{GL}_3(3)$ means the module has a trivial submodule with irreducible 2-dimensional quotient. For $n \le 4$ the only candidates with a non-cyclic Sylow 3-subgroup occur for two classes of $T \cong 3^2$ in $\mathrm{GL}_4(3)$: the extensions of $\mathrm{Alt}(6) \cong \Omega_4^-(3)$ and of $\mathrm{SL}_2(9) = 2.\mathrm{Alt}(6)$ by subgroups of $\mathrm{Out}(\mathrm{Alt}(6))$ and by scalars; $T \cong C_9$, 3^3, 3^4 and 3_+^{1+2} admit no candidates in $\mathrm{GL}_4(3)$. (The Ree groups and $\mathrm{PSU}_3(9)$ need $n \ge 6$, $\mathrm{PSL}_3(4)$ and M_{11} need $n \ge 5$.)

Table 1: Subgroups H of $\mathrm{GL}_2(3)$ with a strongly 3-embedded subgroup, by Sylow 3-subgroup T (one H per $N_{\mathrm{GL}_2(3)}(T)$-class; $|N_H(T)|$, $|O_{3'}(H)|$, whether H is soluble, the dimensions of the composition factors of the natural module).

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N_H(T)\rvert$	$\lvert O_{3'}(H)\rvert$	sol.	factors
$C3$	12	24	`SL(2,3)`	6	8	true	2
$C3$	12	48	`GL(2,3)`	12	8	true	2

Table 2: Subgroups H of $\mathrm{GL}_3(3)$ with a strongly 3-embedded subgroup, by Sylow 3-subgroup T (one H per $N_{\mathrm{GL}_3(3)}(T)$-class; $|N_H(T)|$, $|O_{3'}(H)|$, whether H is soluble, the dimensions of the composition factors of the natural module).

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N_H(T)\rvert$	$\lvert O_{3'}(H)\rvert$	sol.	factors
$C3$	216	24	`SL(2,3)`	6	8	true	1+2
$C3$	216	48	`GL(2,3)`	12	8	true	2+1
$C3$	216	48	`C2 x SL(2,3)`	12	16	true	2+1
$C3$	216	48	`GL(2,3)`	12	8	true	2+1
$C3$	216	96	`C2 x GL(2,3)`	24	16	true	1+2
$C3$	36	12	`A4`	3	4	true	3
$C3$	36	24	`C2 x A4`	6	8	true	3
$C3$	36	24	`S4`	6	4	true	3
$C3$	36	24	`S4`	6	4	true	3
$C3$	36	39	`C13 :  C3`	3	13	true	3
$C3$	36	48	`C2 x S4`	12	8	true	3
$C3$	36	78	`C2 x (C13 :  C3)`	6	26	true	3
3^2	864	none					
3^2	864	none					
3^2	108	none					
3_+^{1+2}	216	none					

Table 3: Subgroups H of $\mathrm{GL}_4(3)$ with a strongly 3-embedded subgroup, by Sylow 3-subgroup T (one H per $N_{\mathrm{GL}_4(3)}(T)$-class; $|N_H(T)|$, $|O_{3'}(H)|$, whether H is soluble, the dimensions of the composition factors of the natural module).

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N$
$C3$	46656	24	`SL(2,3)`	
$C3$	46656	48	`GL(2,3)`	
$C3$	46656	48	`GL(2,3)`	
$C3$	46656	48	`C2 x SL(2,3)`	
$C3$	46656	48	`C2 x SL(2,3)`	

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N$
$C3$	46656	48	GL(2,3)	
$C3$	46656	96	C2 x GL(2,3)	
$C3$	46656	96	C4 x SL(2,3)	
$C3$	46656	96	Q8 : (C3 : C4)	
$C3$	46656	96	C2 x C2 x SL(2,3)	
$C3$	46656	96	C2 x GL(2,3)	
$C3$	46656	96	C2 x GL(2,3)	
$C3$	46656	96	C2 x GL(2,3)	
$C3$	46656	192	SL(2,3) x Q8	
$C3$	46656	192	Q8 : D24	
$C3$	46656	192	C8 x SL(2,3)	
$C3$	46656	192	Q8 : (C3 : C8)	
$C3$	46656	192	Q8 : (C3 : Q8)	
$C3$	46656	192	C4 x GL(2,3)	
$C3$	46656	192	SL(2,3) x D8	
$C3$	46656	192	Q8 : ((C6 x C2) : C2)	
$C3$	46656	192	C2 x C2 x GL(2,3)	
$C3$	46656	384	GL(2,3) x Q8	
$C3$	46656	384	SL(2,3) x QD16	
$C3$	46656	384	GL(2,3) x D8	
$C3$	46656	384	Q8 : ((C3 : Q8) : C2)	
$C3$	46656	384	C8 : GL(2,3)	
$C3$	46656	384	C8 x GL(2,3)	
$C3$	46656	384	Q8 : GL(2,3)	
$C3$	46656	768	GL(2,3) x QD16	
$C3$	648	12	A4	
$C3$	648	24	SL(2,3)	
$C3$	648	24	S4	
$C3$	648	24	S4	
$C3$	648	24	C2 x A4	
$C3$	648	24	C2 x A4	
$C3$	648	24	S4	
$C3$	648	24	C2 x A4	
$C3$	648	24	S4	
$C3$	648	39	C13 : C3	
$C3$	648	48	C2 x S4	
$C3$	648	48	C2 x C2 x A4	
$C3$	648	48	C2 x S4	
$C3$	648	48	C2 x S4	
$C3$	648	48	C2 x S4	
$C3$	648	48	C2 x S4	
$C3$	648	48	C2 x S4	
$C3$	648	48	GL(2,3)	
$C3$	648	60	A5	
$C3$	648	78	C2 x (C13 : C3)	
$C3$	648	78	C2 x (C13 : C3)	

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N$
$C3$	648	78	C2 x (C13 : C3)	
$C3$	648	96	((C2 x C2 x C2) : (C2 x C2)) : C3	
$C3$	648	96	C2 x C2 x S4	
$C3$	648	120	S5	
$C3$	648	120	C2 x A5	
$C3$	648	120	S5	
$C3$	648	156	C2 x C2 x (C13 : C3)	
$C3$	648	192	(C2 x C2 x C2) : S4	
$C3$	648	192	Q8 : S4	
$C3$	648	192	((C4 x C2) : D8) : C3	
$C3$	648	240	C2 x S5	
$C3$	648	384	(C2 x C2 x C2 x C2) : S4	
$C3$	7776	24	SL(2,3)	
$C3$	7776	24	SL(2,3)	
$C3$	7776	48	GL(2,3)	
$C3$	7776	48	C2 . S4 = SL(2,3) . C2	
$C3$	7776	48	((C4 x C2) : C2) : C3	
$C3$	7776	48	C2 x SL(2,3)	
$C3$	7776	48	C2 x SL(2,3)	
$C3$	7776	48	GL(2,3)	
$C3$	7776	48	GL(2,3)	
$C3$	7776	48	GL(2,3)	
$C3$	7776	96	((C2 x C2 x C2) : (C2 x C2)) : C3	
$C3$	7776	96	((C8 x C2) : C2) : C3	
$C3$	7776	96	SL(2,3) : C4	
$C3$	7776	96	(C2 . S4 = SL(2,3) . C2) : C2	
$C3$	7776	96	((C2 x Q8) : C2) : C3	
$C3$	7776	96	((C4 x C2) : C2) : S3	
$C3$	7776	96	((C4 x C2) : C2) : S3	
$C3$	7776	96	C2 x GL(2,3)	
$C3$	7776	96	(C2 x Q8) : S3	
$C3$	7776	96	C2 x GL(2,3)	
$C3$	7776	120	SL(2,5)	
$C3$	7776	192	(Q8 x Q8) : C3	
$C3$	7776	192	((C2 x C2 x C2) : (C2 x C2)) : S3	
$C3$	7776	192	((C4 x C2) : C2) : D12	
$C3$	7776	192	((C2 . S4 = SL(2,3) . C2) : C2) : C2	
$C3$	7776	192	((C2 . S4 = SL(2,3) . C2) : C2) : C2	
$C3$	7776	192	(C2 . S4 = SL(2,3) . C2) : C4	
$C3$	7776	192	SL(2,3) : D8	
$C3$	7776	192	((C8 : C2) : (C2 x C2)) : C3	
$C3$	7776	240	SL(2,5) : C2	
$C3$	7776	240	C2 . S5 = SL(2,5) . C2	
$C3$	7776	240	SL(2,5) : C2	
$C3$	7776	384	(Q8 x Q8) : S3	
$C3$	7776	384	Q8 : GL(2,3)	

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N$
$C3$	7776	384	((C2 . ((C2 x C2 x C2) : C4) = (C4 x C2) . (C4 x C2)) : C2) : C3	
$C3$	7776	384	((C8 x C2) : C2) : D12	
$C3$	7776	480	C8 . A5 = SL(2,5) . C4	
$C3$	7776	480	SL(2,5) : (C2 x C2)	
$C3$	7776	480	(C2 . S5 = SL(2,5) . C2) : C2	
$C3$	7776	768	(Q8 x Q8) : D12	
$C3$	7776	960	(C8 . A5 = SL(2,5) . C4) : C2	
$C3$	7776	1920	C2 . ((C2 x C2 x C2 x C2) : A5)	
$C3$	7776	3840	(C2 . ((C2 x C2 x C2 x C2) : A5)) : C2	
3^2	432	none		
3^2	46656	none		
3^2	648	none		
3^2	648	none		
3^2	1944	none		
3^2	46656	none		
3^2	5832	none		
$C9$	324	none		
3^2	10368	720	SL(2,9)	
3^2	10368	1440	SL(2,9) : C2	
3^2	10368	1440	SL(2,9) : C2	
3^2	10368	2880	SL(2,9) : C4	
3^2	10368	2880	SL(2,9) : (C2 x C2)	
3^2	10368	2880	SL(2,9) : C4	
3^2	10368	5760	SL(2,9) : D8	
3^2	10368	5760	SL(2,9) : Q8	
3^2	10368	5760	GL(2,9)	
3^2	10368	11520	SL(2,9) : QD16	
3^2	2592	none		
3^2	864	360	A6	
3^2	864	720	C2 x A6	
3^2	864	720	A6 . C2	
3^2	864	720	S6	
3^2	864	1440	A6 : C4	
3^2	864	1440	C2 x S6	
3^2	864	1440	C2 x (A6 . C2)	
3^2	864	2880	(C2 x (A6 . C2)) : C2	
3^3	5832	none		
3^{1+2}_+	3888	none		
3^{1+2}_+	2916	none		
3^3	7776	none		
3^{1+2}_+	1944	none		
3^3	3888	none		
3^{1+2}_+	3888	none		
3^3	7776	none		
3^3	606528	none		
3^{1+2}_+	23328	none		

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N$
3_+^{1+2}	972	none		
3^3	5832	none		
3_+^{1+2}	1944	none		
3^3	11664	none		
3_+^{1+2}	1944	none		
3_+^{1+2}	972	none		
3^3	606528	none		
3_+^{1+2}	1944	none		
3^4	186624	none		

$\mathrm{GL}_5(3)$, $T \cong 3^2$. For $n = 5$ the subgroup lattice is out of reach, but for $T \cong 3^2$ the candidates can be constructed directly: $O^{3'}(H/O_{3'}(H))$ is Alt(6), $\mathrm{PSL}_3(4)$ or M_{11}, and $7 \nmid |\mathrm{GL}_5(3)|$ excludes $\mathrm{PSL}_3(4)$; hence the perfect residuum Y of H is Alt(6), $\mathrm{SL}_2(9)$ or M_{11} (M_{11} has trivial Schur multiplier), and H lies between Y and $N_{\mathrm{GL}_5(3)}(Y)$ by the Frattini argument. Alt(6) and $\mathrm{SL}_2(9)$ have no faithful $\mathbb{F}_3$-module of dimension 5 other than $4 + 1$, so they come from the $\mathrm{GL}_4(3)$ table; M_{11} has two dual irreducible 5-dimensional modules (constituents of the module induced from a 5-dimensional constituent of the permutation module of $\mathrm{PSL}_2(11) < M_{11}$ on 11 points), giving two further classes of T. The script is `scripts/gl5direct.g`; the four classes of $T \cong 3^2$ in $\mathrm{GL}_5(3)$ with candidates carry 34, 27, 2 and 2 of them.

Table 4: Subgroups H of $\mathrm{GL}_5(3)$ with a strongly 3-embedded subgroup, by Sylow 3-subgroup T (one H per $N_{\mathrm{GL}_5(3)}(T)$-class; $|N_H(T)|$, $|O_{3'}(H)|$, whether H is soluble, the dimensions of the composition factors of the natural module).

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N_H(T)\rvert$	$\lvert O_{3'}(H)\rvert$	sol.	factors
3^2	1679616	720	`SL(2,9)`	72	2	false	1+4
3^2	1679616	23040	`C2 x (SL(2,9) : QD16)`	2304	16	false	4+1
3^2	1679616	1440	`SL(2,9) : C2`	144	2	false	4+1
3^2	1679616	1440	`SL(2,9) : C2`	144	2	false	1+4
3^2	1679616	1440	`SL(2,9) : C2`	144	4	false	1+4
3^2	1679616	1440	`C2 x SL(2,9)`	144	4	false	1+4
3^2	1679616	1440	`SL(2,9) : C2`	144	4	false	1+4
3^2	1679616	2880	`SL(2,9) : (C2 x C2)`	288	4	false	4+1
3^2	1679616	2880	`SL(2,9) : C4`	288	8	false	1+4
3^2	1679616	2880	`SL(2,9) : C4`	288	4	false	4+1
3^2	1679616	2880	`SL(2,9) : (C2 x C2)`	288	4	false	1+4
3^2	1679616	2880	`SL(2,9) : C4`	288	8	false	1+4
3^2	1679616	2880	`C2 x (SL(2,9) : C2)`	288	4	false	4+1
3^2	1679616	2880	`SL(2,9) : (C2 x C2)`	288	4	false	1+4
3^2	1679616	2880	`C2 x (SL(2,9) : C2)`	288	8	false	1+4
3^2	1679616	2880	`SL(2,9) : C4`	288	4	false	4+1
3^2	1679616	5760	`SL(2,9) : Q8`	576	8	false	1+4
3^2	1679616	5760	`C2 x (SL(2,9) : C4)`	576	8	false	4+1

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N_H(T)\rvert$	$\lvert O_{3'}(H)\rvert$	sol.	factors
3^2	1679616	5760	C2 x (SL(2,9) : C4)	576	16	false	4+1
3^2	1679616	5760	SL(2,9) : Q8	576	8	false	4+1
3^2	1679616	5760	C2 x (SL(2,9) : (C2 x C2))	576	8	false	4+1
3^2	1679616	5760	SL(2,9) : D8	576	8	false	1+4
3^2	1679616	5760	SL(2,9) : D8	576	8	false	1+4
3^2	1679616	5760	GL(2,9)	576	8	false	4+1
3^2	1679616	5760	GL(2,9)	576	8	false	1+4
3^2	1679616	5760	SL(2,9) : Q8	576	8	false	1+4
3^2	1679616	5760	SL(2,9) : D8	576	8	false	4+1
3^2	1679616	11520	C2 x (SL(2,9) : Q8)	1152	16	false	4+1
3^2	1679616	11520	SL(2,9) : QD16	1152	8	false	1+4
3^2	1679616	11520	C2 x (SL(2,9) : D8)	1152	16	false	4+1
3^2	1679616	11520	SL(2,9) : QD16	1152	8	false	4+1
3^2	1679616	11520	C2 x GL(2,9)	1152	16	false	1+4
3^2	1679616	11520	SL(2,9) : QD16	1152	8	false	1+4
3^2	1679616	11520	SL(2,9) : QD16	1152	8	false	4+1
3^2	15552	360	A6	36	1	false	1+4
3^2	15552	5760	C2 x ((C2 x (A6 . C2)) : C2)	576	4	false	4+1
3^2	15552	720	S6	72	1	false	1+4
3^2	15552	720	C2 x A6	72	2	false	1+4
3^2	15552	720	A6 . C2	72	1	false	4+1
3^2	15552	720	A6 . C2	72	1	false	4+1
3^2	15552	720	C2 x A6	72	2	false	1+4
3^2	15552	720	S6	72	1	false	4+1
3^2	15552	720	C2 x A6	72	2	false	1+4
3^2	15552	1440	C2 x (A6 . C2)	144	2	false	1+4
3^2	15552	1440	C2 x (A6 . C2)	144	2	false	1+4
3^2	15552	1440	C2 x S6	144	2	false	4+1
3^2	15552	1440	C2 x S6	144	2	false	1+4
3^2	15552	1440	C2 x C2 x A6	144	4	false	1+4
3^2	15552	1440	C2 x S6	144	2	false	4+1
3^2	15552	1440	C2 x (A6 . C2)	144	2	false	1+4
3^2	15552	1440	A6 : C4	144	2	false	4+1
3^2	15552	1440	C2 x S6	144	2	false	1+4
3^2	15552	1440	C2 x (A6 . C2)	144	2	false	1+4
3^2	15552	1440	A6 : C4	144	2	false	1+4
3^2	15552	2880	C2 x (A6 : C4)	288	4	false	1+4
3^2	15552	2880	(A6 : C4) : C2	288	2	false	4+1
3^2	15552	2880	(C2 x (A6 . C2)) : C2	288	2	false	4+1
3^2	15552	2880	C2 x C2 x S6	288	4	false	4+1
3^2	15552	2880	C2 x C2 x (A6 . C2)	288	4	false	4+1
3^2	15552	2880	(A6 : C4) : C2	288	2	false	4+1
3^2	15552	2880	(A6 : C4) : C2	288	2	false	4+1
3^2	2592	7920	M11	144	1	false	5
3^2	2592	15840	C2 x M11	288	2	false	5
3^2	2592	7920	M11	144	1	false	5

T	$\lvert N(T)\rvert$	$\lvert H\rvert$	H	$\lvert N_H(T)\rvert$	$\lvert O_{3'}(H)\rvert$	sol.	factors
3^2	2592	15840	`C2 x M11`	288	2	false	5

Completeness of the automizer candidates. For an essential subgroup E the candidates for $\mathrm{Out}_{\mathcal{F}}(E)$ are complete in the following cases: $|\mathrm{Out}(E)| \le 5\cdot 10^5$ (or $\mathrm{Out}(E)$ soluble of order at most 10^9), where the subgroup lattice of $\mathrm{Out}(E)$ is used; E elementary abelian of rank at most 4 (the tables above); E elementary abelian of rank 5 with $\mathrm{Out}_S(E) \cong 3^2$ (the table above); and every E with $\mathrm{Out}_S(E)$ neither cyclic nor quaternion (then $\mathrm{Out}_{\mathcal{F}}(E)$ is not 3-soluble and the perfect subgroup method of the package is complete). In the remaining case, an elementary abelian E of rank 5, 6 or 7 with $\mathrm{Out}_S(E)$ cyclic and $|\mathrm{Out}(E)| = |\mathrm{GL}_n(3)|$ beyond the lattice bound, the non-3-soluble candidates are complete but the 3-soluble ones ($H = O_{3'}(H) \rtimes N_H(T)$ with T acting faithfully on the $3'$-group $O_{3'}(H) \le \mathrm{GL}_n(3)$) are only found inside the monomial groups of the T-invariant frames. Results for groups whose only proto-essential subgroups of this kind are elementary abelian of rank at least 5 with cyclic $\mathrm{Out}_S(E)$ are therefore marked "complete up to the soluble candidates" below; such groups are, by the ordering of Stage 2, treated last. (Note that for $\mathrm{Out}_S(E) \cong C_9$ or C_{27} the faithful action of $\Omega_1(T)$ on $O_{3'}(H)$ needs an elementary abelian 2-section of rank at least 6 or an element of order 19 or 37, which already restricts the possibilities severely.)

Use of the tables. In Stage 1a an elementary abelian candidate $E \cong 3^n$ is discarded unless $\mathrm{Aut}_S(E)$ is conjugate in $\mathrm{GL}_n(3)$ to a class with candidates ($n \le 4$, and $n = 5$ for $\mathrm{Out}_S(E) \cong 3^2$), and in Stage 2 the automizer candidates for such E are taken from the tables (transported along the conjugating matrix) instead of being enumerated anew for each group. This matters for correctness as well as for speed: for $|\mathrm{GL}_n(3)|$ beyond the lattice bound the package's own enumeration of the soluble candidates for cyclic $\mathrm{Out}_S(E)$ (the frame method) is not complete, e.g. it misses the irreducible $13{:}3 \le \mathrm{GL}_3(3) \le \mathrm{GL}_4(3)$.

6 Results group by group

Each subsection treats one group S and lists the saturated fusion systems $\mathcal{F}$ on S with $O_3(\mathcal{F}) = 1$ and $O^3(\mathcal{F}) = \mathcal{F}$ found by `AllFusionSystems`, up to isomorphism. For each $\mathcal{F}$ we give $\mathrm{Out}_{\mathcal{F}}(S)$, the essential subgroups E up to $\mathcal{F}$-conjugacy (with $|N_S(E) : E|$) and $\mathrm{Out}_{\mathcal{F}}(E)$, in the style of the tables of [1]. Identifications with fusion systems of finite groups are proved by `IsIsomorphicFusionSystems` against the fusion system of the group when the group is available in GAP, and are otherwise indicated as expectations. The groups are treated in the order of difficulty (the order of the largest elementary abelian proto-essential subgroup).

6.1 The Sylow 3-subgroup of $\mathrm{PSL}_3(q)$, $3^4 \parallel q-1$ (maximal class): `SmallGroup(6561,2221)`

$S = \big(\{(x,y,z) \in C_{81}^3 : xyz = 1\} \rtimes C_3\big)/C_3$ is `SmallGroup(6561,2221)` (verified with `IsomorphismGroups`); it has maximal class, $|\mathrm{Z}(S)| = 3$, exponent 81 and an abelian maximal subgroup $A \cong C_{81} \times C_{81}$; it is the Sylow 3-subgroup of $\mathrm{PSL}_3(q)$ for $3^4 \parallel q-1$ (e.g. $q = 163$) and of $\mathrm{PSU}_3(q)$ for $3^4 \parallel q+1$ (e.g. $q = 647$), and $|\mathrm{Aut}(S)| = 2^2 \cdot 3^{13}$. The proto-essential subgroups are three $\mathrm{Aut}(S)$-classes of elementary abelian subgroups $E \cong 3^2$ with $|N_S(E) : E| = 3$ (so $\mathrm{Out}_{\mathcal{F}}(E) \in \{\mathrm{SL}_2(3), \mathrm{GL}_2(3)\}$ by the $\mathrm{GL}_2(3)$ table) and three $\mathrm{Aut}(S)$-classes of subgroups 3_+^{1+2} with $|N_S(E) : E| = 3$. There are five Borel subgroups, $|\mathrm{Out}_{\mathcal{F}}(S)| \in \{1,2,2,2,4\}$, and the computation takes 17 seconds. Result: exactly 7 fusion systems.

$\mathcal{F}$	$\mathrm{Out}_{\mathcal{F}}(S)$	essential subgroups E, $\lvert N_S(E):E\rvert$, $\mathrm{Out}_{\mathcal{F}}(E)$	remarks
1	C_2	3^2, 3, $\mathrm{SL}_2(3)$; 3^2, 3, $\mathrm{SL}_2(3)$; 3^2, 3, $\mathrm{SL}_2(3)$	see text
2	C_2	3^2, 3, $\mathrm{SL}_2(3)$; 3^2, 3, $\mathrm{SL}_2(3)$	expected $\mathrm{PSL}_3(q)$
3	C_2	3^2, 3, $\mathrm{SL}_2(3)$	see text
4	C_2^2	3^{1+2}_+, 3, $\mathrm{GL}_2(3)$; 3^2, 3, $\mathrm{SL}_2(3)$	see text
5	C_2^2	3^2, 3, $\mathrm{GL}_2(3)$; 3^2, 3, $\mathrm{SL}_2(3)$	see text
6	C_2^2	3^2, 3, $\mathrm{GL}_2(3)$	see text
7	C_2^2	3^2, 3, $\mathrm{SL}_2(3)$	see text

[Identification in progress: the fusion system of $\mathrm{PSL}_3(163)$ (as a permutation group on the 26 733 points of the projective plane) and of its extension by the graph automorphism are being computed for comparison with `IsIsomorphicFusionSystems`. Saturated fusion systems on 3-groups of maximal class were classified by Díaz, Ruiz and Viruel (*All p-local finite groups of rank two for odd prime p*, Trans. AMS 2007); the seven systems above should be compared with their table for the group $B(3,8;0,0,0)$, which contains exotic systems.] The run of `AllFusionSystems` on `SmallGroup(6561,2221)` from the library (Stage 2 of the pipeline, with the candidates of Stage 1a) gives the same seven systems in 19 seconds.

6.2 The other maximal class group with $\lvert\mathrm{Z}(S)\rvert = 3$ and exponent 81: `SmallGroup(6561,2225)`

$S =$ `SmallGroup(6561,2225)` has maximal class, $\lvert\mathrm{Z}(S)\rvert = 3$, exponent 81, 305 conjugacy classes, no abelian maximal subgroup and $\lvert\mathrm{Aut}(S)\rvert = 2 \cdot 3^{12}$. Its only proto-essential subgroup is an elementary abelian $E \cong 3^2$ with $\lvert N_S(E):E\rvert = 3$ (one S-class). There is exactly one fusion system:

$\mathcal{F}$	$\mathrm{Out}_{\mathcal{F}}(S)$	essential subgroups	remarks
1	C_2	3^2, 3, $\mathrm{SL}_2(3)$	[to be identified]

(7 seconds.)

6.3 Groups of class 6 with two generators carrying no fusion system

Five 2-generator groups of class 6 and exponent 27 survive Stages 1a and 1b with proto-essential subgroups $E \cong 3^3$, $\lvert N_S(E):E\rvert = 3$, only, but `AllFusionSystems` finds no saturated fusion system with $O_3(\mathcal{F}) = 1$ and $O^3(\mathcal{F}) = \mathcal{F}$ on any of them (a few seconds each):

S	$\lvert\mathrm{Z}(S)\rvert$	$\lvert S'\rvert$	$\lvert\mathrm{Aut}(S)\rvert$	abelian max. subgroup	proto-essential classes
`SmallGroup(6561,2025)`	9	3^6	$2 \cdot 3^{12}$	no	1
`SmallGroup(6561,2029)`	3	3^6	$2 \cdot 3^{12}$	no	1
`SmallGroup(6561,2193)`	9	3^5	$4 \cdot 3^{12}$	yes	1
`SmallGroup(6561,2021)`	3	3^6	$2 \cdot 3^{11}$	no	3
`SmallGroup(6561,2205)`	3	3^5	$2 \cdot 3^{12}$	no	3

In each case the Borel subgroup $\mathrm{Aut}_{\mathcal{F}}(S) = \mathrm{Inn}(S)$ is excluded by the focal subgroup and for $\lvert\mathrm{Out}_{\mathcal{F}}(S)\rvert = 2$ no choice of automizers ($\mathrm{Out}_{\mathcal{F}}(E) \leq \mathrm{GL}_3(3)$ from the table of Section 5) gives a saturated system.

6.4 Groups whose proto-essential subgroups are 3^4 with $|N_S(E) : E| = 3$

Among the first 92 groups passing Stage 1b, six have proto-essential subgroups $E \cong 3^4$, $|N_S(E) : E| = 3$ (so $\mathrm{Out}_{\mathcal{F}}(E)$ is one of the 30, 33 or 43 candidates of the three classes of C_3 in the $\mathrm{GL}_4(3)$ table) and nothing else: `SmallGroup(6561,`n`)` for $n = 1111$ (3 classes of E), 1115 (3), 1167 (5), 1176 (9), 1225 (9), 1229 (9). None of them carries a saturated fusion system with $O_3(\mathcal{F}) = 1$ and $O^3(\mathcal{F}) = \mathcal{F}$ (44 to 143 seconds each).

References

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[3] C. Broto, N. Castellana, J. Grodal, R. Levi, B. Oliver, *Subgroup families controlling p-local finite groups*, Proc. LMS 91 (2005).